



Term Evaluation Theory (Even) Semester Regular Examination February 2026

Roll no.....

Name of the Course: M. Tech. (CSE)
Semester: II
Name of the Paper: Reinforcement Learning
Paper Code: MCS-203
Time: 1.5-hour

Maximum Marks: 50

Note:

- Answer all the questions by choosing any one of the sub-questions
- Each question carries 10 marks.

Q1. (10 Marks)

- a. Consider Markov decision process with two states: $S = \{s_1, s_2\}$ and two actions: $A = \{a_1, a_2\}$. The transition probabilities and rewards are given below:

From state s_1 :

Action	Next state	Probability	Reward
a_1	s_1	0.5	5
	s_2	0.5	10
a_2	s_1	0.7	3
	s_2	0.3	8

From state s_2 :

Action	Next state	Probability	Reward
a_1	s_2	1.0	4
a_2	s_1	0.4	6
	s_2	0.6	2

And discount factor $\gamma = 0.9$.

Compute the following:

- Write the Bellman optimality equations.
- Compute the optimal value function $V^*(s_1)$ and $V^*(s_2)$.
- Determine the optimal policy.

CO1

OR

- b. What is a Markov Decision Process (MDP)? Describe its components, Bellman equations, and applications.

CO1

Q2. (10 Marks)

- a. A system can be in three states $S = \{A, B, C\}$. The state transition probability matrix is given as:

$$P = \begin{bmatrix} 0.6 & 0.3 & 0.1 \\ 0.2 & 0.5 & 0.3 \\ 0.4 & 0.1 & 0.5 \end{bmatrix}$$

Where $P_{ij} = P(X_{t+1} = j | X_t = i)$.

Answer the following:

- Verify whether the process satisfies the Markov property.
- Find the probability that the system is in state C after two steps, given that it starts in state A.
- Find $P(X_2 = B | X_0 = A)$.

CO1

OR

- b. Compare Supervised Learning, Unsupervised Learning, and Reinforcement Learning. Why is Reinforcement Learning considered fundamentally different from Supervised and Unsupervised Learning?

CO1



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Q3. (10 Marks)

- a. An agent follows a fixed policy π in an episodic environment. The following three episodes are generated. The discount factor is: $\gamma \doteq 1$. Each episode ends in a terminal state.
Episodes and Rewards:

Episode 1:

$A \rightarrow B \rightarrow C \rightarrow \text{Terminal}$

Rewards: 2, 3, 5

Episode 2:

$A \rightarrow C \rightarrow \text{Terminal}$

Rewards: 4, 6

Episode 3:

$B \rightarrow C \rightarrow \text{Terminal}$

Rewards: 1, 5.

Then compute the following:

- i. Using First-Visit Monte Carlo Prediction, estimate: $V(A)$, $V(B)$, $V(C)$.
ii. Show all calculations clearly.

CO2

OR

- b. Explain Bellman Equations for Prediction in Reinforcement Learning.

CO1

Q4. (10 Marks)

- a. Explain the comparative analysis between first-visit and every-visit Monte Carlo prediction in reinforcement learning.

CO2

OR

- b. Explain dynamic programming for solving reinforcement learning problems with a suitable example and what are the advantages and limitations of dynamic programming.

CO2

Q5. (10 Marks)

- a. Explain on-policy and off-policy learning in reinforcement learning with a suitable example.

CO1

OR

- c. Perform a comparative analysis between dynamic programming and Monte Carlo methods in reinforcement learning.

CO2